Upper Division Tutoring Program Topic Review

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Group Theory

Section I: Definitions and Theorems Review

Groups

Definition 1 (group)**.** A *group* G is a set G equipped with a binary operation $\cdot\colon G \times G \to G$ satisfying the following axioms.

- Closure: for each $g_1, g_2 \in G$, we have $g_1 \cdot g_2 \in G$.
- Associativity: for each $g_1, g_2, g_3 \in G$, we have $(g_1 \cdot g_2) \cdot g_3 = g_1 \cdot (g_2 \cdot g_3)$.
- Identity: there is some $e \in G$ such that $g \cdot e = e \cdot g = g$ for each $g \in G$.
- Inverses: for each $g \in G$, there is some element $g^{-1} \in G$ such that $g \cdot g^{-1} = g^{-1} \cdot g = e$.

In the future, e will always denote the identity of the group G. The *order* of G is the size of G as a set.

Definition 2 (abelian)**.** A group G is *abelian* or *commutative* if and only if $g_1 \cdot g_2 = g_2 \cdot g_1$ for each $g_1, g_2 \in G$.

Example 1. Let n be a positive integer. Then the set S_n of permutations of the set $\{1, 2, \ldots, n\}$ is a group, where the operation is composition. If $n \geq 3$, the group S_n is not abelian.

Theorem 1 (Fundamental Theorem of Finitely Generated Abelian Groups)**.** Every finitely generated abelian group G is isomorphic to

$$G \cong \mathbb{Z}^r \times \mathbb{Z}/p_1^{k_1}\mathbb{Z} \times \cdots \times \mathbb{Z}/p_m^{k_m}\mathbb{Z},$$

where $r \geq 0$, the p_i are primes, and $k_i \geq 1$. The decomposition is unique up to reordering the prime-power factors.

Subgroups

Definition 3 (subgroup)**.** Fix a group G. A *subgroup* H is a subset $H \subseteq G$ which is a group under the same operation. Rigorously, we ask for H to satisfy the following properties.

- Identity: we have $e \in H$, where e is the identity.
- Closure: for each $h_1, h_2 \in H$, we have $h_1 \cdot h_2 \in H$.
- Inverses: for each $h \in H$, we have $h^{-1} \in H$.

Definition 4 (normal)**.** Fix a subgroup H of a group G. Then H is a *normal subgroup* if and only if $g \cdot h \cdot g^{-1} \in H$ for each $g \in G$ and $h \in H$.

Definition 5 (coset)**.** Fix a subgroup H of a group G. Then a *left coset* of H is a set of the form

$$g \cdot H \coloneqq \{g \cdot h : h \in H\}.$$

We let G/H denote the set of left cosets. There is a symmetric notion of right cosets $H \cdot g$, and the set of right cosets is denoted $H\backslash G$. The *index* of H in G, denoted $[G : H]$, is the number of (left) cosets.

Theorem 2 (Lagrange)**.** Let H be a subgroup of a finite group G. Then $|G| = |G/H| \cdot |H|$. In particular, $|H|$ divides $|G|$.

Definition 6 (quotient group)**.** Fix a normal subgroup H of a group G. Then the set of cosets G/H forms a group with the operation $g_1H \cdot g_2H \coloneqq (g_1g_2H)$.

Example 2. Let n be a positive integer. Then the set $n\mathbb{Z}$ of integer multiples of n is a normal subgroup of $\mathbb{Z}$. The group $\mathbb{Z}/n\mathbb{Z}$ is then an abelian group with n elements.

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Homomorphisms

Definition 7 (homomorphism)**.** A *homomorphism* is a function $\varphi\colon G \to G'$ of groups such that

$$\varphi(g_1 \cdot g_2) = \varphi(g_1) \cdot \varphi(g_2)$$

for each $g_1, g_2 \in G$. If φ is a bijection, we call φ an *isomorphism*.

Definition 8 (kernel)**.** Fix a homomorphism $\varphi\colon G \to G'$ of groups, and let e' be the identity of G'. Then the *kernel of* φ is the set

$$\ker \varphi \coloneqq \{g \in G : \varphi(g) = e'\}.$$

It turns out that $\ker \varphi$ is a normal subgroup of G.

Definition 9 (image)**.** Fix a homomorphism $\varphi\colon G \to G'$ of groups. Then the *image of* φ is the set

$$\operatorname{im} \varphi \coloneqq \{\varphi(g) : g \in G\}.$$

It turns out that $\operatorname{im} \varphi$ is a subgroup of G'.

Theorem 3 (homomorphism)**.** Fix a homomorphism $\varphi\colon G \to G'$ of groups. Then the map

$$\overline{\varphi}\colon \frac{G}{\ker \varphi} \to \operatorname{im} \varphi$$

defined by $\overline{\varphi}(g(\ker \varphi)) \coloneqq \varphi(g)$ is an isomorphism of groups.

Group Actions

Definition 10 (group action)**.** Fix a group G and a set S. A *group action of* G *on* S is an operation $\cdot\colon G \times S \to S$ satisfying the following axioms.

- Associativity: we have $(g_1 \cdot g_2) \cdot s = g_1 \cdot (g_2 \cdot s)$ for each $g_1, g_2 \in G$ and $s \in S$.
- Identity: we have $e \cdot s = s$ for each $s \in S$.

Remark. It is occasionally illuminating to view a group action of a group G on a set S as a group homomorphism $G \to \operatorname{Sym}(S)$, where $\operatorname{Sym}(S)$ is the group of bijective functions $S \to S$ whose operation is given by composition.

Example 3. Fix a positive integer n. Then the group S_n acts on the set $\{1, 2, \ldots, n\}$ by permutations: for $\sigma \in S_n$ and $k \in \{1, 2, \ldots, n\}$, we have $\sigma \cdot k \coloneqq \sigma(k)$.

Definition 11 (stabilizer)**.** Fix a group G acting on a set S. Given $s \in S$, the *stabilizer* of $s \in S$ is the set

$$\operatorname{Stab}(s) \coloneqq \{g \in G : g \cdot s = s\}.$$

It turns out that $\operatorname{Stab}(s)$ is a subgroup of G.

Remark. Fix a group G acting on a set S. Given $s \in S$ and $g \in G$, one can show that $g \cdot \operatorname{Stab}(s) \cdot g^{-1} = \operatorname{Stab}(g \cdot s)$.

Definition 12 (orbit)**.** Fix a group G acting on a set S. Given $s \in S$ the *orbit* of $s \in S$ is the set

$$\operatorname{Orb}(s) \coloneqq \{g \cdot s : g \in G\}.$$

Theorem 4 (orbit–stabilizer)**.** Fix a group G acting on a set S. For any $s \in S$, we have $|\operatorname{Orb}(s)| \cdot |\operatorname{Stab}(s)| = |G|$.

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Section 2: Problems

1. Consider the group S_3 of permutations on the set $\{1, 2, 3\}$. Let $H \subseteq S_3$ be a normal subgroup containing the transposition $(12) \in H$.

   (a) Compute $(13)(12)(13)^{-1}$ to show that $(23) \in H$. Using a similar computation, show that $(13) \in H$.

   (b) Compute $(12)(23)$ to show that $(123) \in H$. Using a similar computation, show that $(132) \in H$.

   (c) Conclude that $H = S_3$.

   As an aside, we remark that a similar argument shows that any normal subgroup $H \subseteq S_n$ containing a transposition must be all of S_n.

2. Let G be a group in which $x^2 = e$ for all $x \in G$. Prove that G is abelian.

3. Let G be a group, and suppose that two subgroups $H, K \subseteq G$ have coprime order. Show that $H \cap K = \{e\}$.

4. Define the group homomorphism $\varphi\colon (\mathbb{Z} \times \mathbb{Z}) \to \mathbb{Z}$ by $\varphi(a, b) \coloneqq 2a - 3b$.

   (a) Compute $\ker \varphi$ as a subgroup of $\mathbb{Z} \times \mathbb{Z}$ and $\operatorname{im} \varphi$ as a subgroup of $\mathbb{Z}$.

   (b) Show that $\ker \varphi$ is isomorphic to $\mathbb{Z}$.

5. We count the number of homomorphisms between two groups.

   (a) Compute the number of homomorphisms $\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$.

   (b) Compute the number of homomorphisms $\mathbb{Z} \to S_3$.

   (c) Compute the number of isomorphisms $\mathbb{Z} \to \mathbb{Z}$.

   (d) Compute the number of homomorphisms $\mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}$.

   (e) Compute the number of homomorphisms $\mathbb{Z}/2\mathbb{Z} \to S_3$.

6. Fix a group G. Define the *center* of G to be the set

$$Z(G) \coloneqq \{g \in G : gx = xg \text{ for all } x \in G\}.$$

   (a) Show that $Z(G)$ is a normal subgroup of G.

   (b) If $H \subseteq G$ is a normal subgroup of size 2, then $H \subseteq Z(G)$.

7. Fix a subgroup H of a group G such that G/H has exactly two elements. In other words, there are exactly two left cosets of H.

   (a) Show that the two left cosets in G/H are H and $G \setminus H$. Similarly, show that the two right cosets in G/H are H and $G \setminus H$.

   (b) For each $g \in G \setminus H$, show that $g \cdot H = G \setminus H$ and $H \cdot g = G \setminus H$.

   (c) Show that H is a normal subgroup of G.

8. Let a group G act on a set S. Define the set

$$K \coloneqq \{g \in G : g \cdot s = s \text{ for all } s \in S\}.$$

   In other words, K is the subset of G fixing every element of S. Show that K is a normal subgroup of G.

9. A subgroup T of a group W is called characteristic if $\varphi(T) \subset T$ for all automorphisms, φ, of W. Prove that:

- M characteristic in G imples that $M \triangleleft G$.
- M, N characteristic in G imples that MN is characteristic in G

10. Let
$$D_3 = \langle r, s \mid r^3 = s^2 = e,\ srs = r^{-1} \rangle,$$
where r is a rotation and s is a reflection. Draw the Cayley diagram of D_3 with respect to the generators r and s.

11. Let G be an abelian group of order
$$|G| = 144 = 2^4 \cdot 3^2.$$
(a) How many abelian groups of order 144 are there, up to isomorphism?
(b) How many of these groups contain an element of order 16?
(c) How many of these groups have no element of order 8?

Bonus. Consider the following 4×4 sliding puzzle, where the empty square is in the upper-left corner. A legal move slides a tile adjacent to the empty square into that square.

	2	7	11
4	15	1	9
13	5	10	3
8	12	6	14

Can this puzzle be solved so that the tiles are in increasing order, with the empty square in the upper-left corner? Justify your answer.